

Name: _____ Class: _____

SYDNEY TECHNICAL HIGH SCHOOL



Year 12 Mathematics Extension 2 HSC Assessment Task 2 June 2005

Instructions:

- Begin each question on a new page.
- Write only on the front of each page. Single column only.
- Staple these questions to the front of your answers.
- Full marks may not be awarded for careless or incomplete work.
- Marks indicated in each question are a guide only and may change slightly during the marking process.

Time allowed: ***70 mins***

Q1	Q2	Q3	TOTAL

Question 1.

- a) Factorise $x^3 + x^2 - x + 15$ over the complex field given that $1-2i$ is a zero. 3

b) Find i) $\int \frac{e^{\tan x}}{\cos^2 x} dx$ 1

ii) $\int \sec^6 x \, dx$ 3

iii) $\int \frac{x-2}{\sqrt{x^2-9}} dx$. 3

- c) $P(2p, \frac{2}{p})$ and $Q(2q, \frac{2}{q})$ where $p, q > 0$ are two distinct points on the hyperbola $xy = 4$. 7

- i) Show that the equation of the tangent to $xy = 4$ at $P(2p, \frac{2}{p})$ is

$$x + p^2 y - 4p = 0.$$

- ii) If the tangents at $P(2p, \frac{2}{p})$ and $Q(2q, \frac{2}{q})$ intersect at T , find the coordinates of T .

- iii) If the tangent at Q passes through the point $(2p, 0)$ show that $p = 2q$.

- iv) Find the equation of the locus of T .

Question 2.

a) Find $\int \sin^2 2x \cos x \, dx$ 4

b) Find $\int \frac{\ln x}{x^2} dx$ using integration by parts. 3

c) Find $\int_0^{\frac{1}{2}} \sqrt{1-x^2} \, dx$ using a suitable substitution 3

d) The cubic equation $x^3 + kx + 1 = 0$, where k is a constant, 7
has roots p, q and r .

For each non negative integer n , $S_n = p^n + q^n + r^n$.

i) State the value of S_1 .

ii) Express S_2 in terms of k .

iii) Show that for all n , $S_{n+3} + kS_{n+1} + S_n = 0$.

iv) Hence, or otherwise, express $p^5 + q^5 + r^5$ in terms of k .

Question 3.

a) Evaluate $\int_4^9 \frac{\sqrt{x}}{x-1} dx$ using the substitution $u^2 = x$ 5

b) If α , β and δ are the roots of $x^3 + 2x - 1 = 0$, form the equation whose roots are 5

i) α , $-\alpha$, β , $-\beta$, δ and $-\delta$

ii) $\beta + \delta - 2\alpha$, $\delta + \alpha - 2\beta$ and $\alpha + \beta - 2\delta$.

c) A polynomial $P(x)$ is given by $P(x) = x^5 - 1$. 6

Let α ($\alpha \neq 1$) be that complex root of $P(x)$ which has the smallest positive argument.

i) Show that $P(x) = (x-1)(1+x+x^2+x^3+x^4)$

ii) Show that $1 + \alpha + \alpha^2 + \alpha^3 + \alpha^4 = 0$

iii) If $p = \alpha + \alpha^4$ and $q = \alpha^2 + \alpha^3$ find the values of $p+q$ and pq .

iv) Show that $p = \frac{-1+\sqrt{5}}{2}$ and $q = \frac{-1-\sqrt{5}}{2}$. Justify your answer.

End of paper.

Question 1

a) $x^3 + x^2 - x + 15$
 $= (x - (1-2i))(x - (1+2i))(x - ?)$
 $= (x^2 - 2x + 5)(x + 3)$
 $\therefore = (x - (1-2i))(x - (1+2i))(x + 3)$

b) i) $\int \frac{e^{\tan x}}{\cos^2 x} dx$
 $= \int e^{\tan x} \cdot \sec^2 x dx$
 $= e^{\tan x} + c$

ii) $\int \sec^6 x dx$
 $= \int \sec^4 x \sec^2 x dx$
 $= \int (1 + \tan^2 x)^2 \sec^2 x dx \quad (\text{put } u = \tan x)$
 $= \int (1 + 2u^2 + u^4) du$
 $= u + \frac{2u^3}{3} + \frac{u^5}{5} + c$
 $= \tan x + \frac{2 \tan^3 x}{3} + \frac{\tan^5 x}{5} + c$

iii) $\int \frac{x-2}{\sqrt{x^2-9}} dx = \int \frac{x dx}{\sqrt{x^2-9}} - 2 \int \frac{dx}{\sqrt{x^2-9}}$
 $= \sqrt{x^2-9} - 2 \ln(x + \sqrt{x^2-9}) + c$

(3)

c) i) $y = \frac{4}{x} \rightarrow y' = -\frac{4}{x^2}$

when $x = 2p$, $m = -\frac{1}{p^2}$
 Equ of Tangent at $(2p, \frac{2}{p})$ is

$y - \frac{2}{p} = -\frac{1}{p^2}(x - 2p)$

ie $yp^2 - 2p = -x + 2p$

ie $x + yp^2 - 4p = 0$

ii) $x + yp^2 - 4p = 0$ tangent at P — (1)

$x + yq^2 - 4q = 0$ — tangent at Q — (2)

$\therefore y(p^2 q) = 4(p - q)$

ie $y = \frac{4}{p+q}$

$\therefore x = 4p - p^2 \left(\frac{4}{p+q} \right)$
 $= \frac{4pq}{p+q}$

$\therefore T$ is $\left(\frac{4pq}{p+q}, \frac{4}{p+q} \right)$ — (3)

iii) $(2p, 0)$ lies on (2)

$\therefore 2p + 0 = 4q$

$\therefore p = 2q$ — (4)

iv) From (3), $x = y \cdot pq$

but using (4), $pq = 2q^2$ and

putting this in (3), $y = \frac{4}{3q} \Rightarrow q = \frac{4}{3y}$

$\therefore x = y \cdot 2 \left(\frac{4}{3y} \right)^2$
 $= \frac{32}{9y}$

ie $xy = \frac{32}{9}$

(7)

Question 2

a) $\int \sin^2 x \cdot \cos x dx$

$$= \int (2 \sin x \cos x)^2 \cdot \cos x dx$$

$$= 4 \int \sin^2 x \cos^2 x \cdot \cos x dx$$

$$= 4 \int \sin^2 x (1 - \sin^2 x) \cdot \cos x dx$$

put $u = \sin x$

$$= 4 \int u^2 (1 - u^2) du$$

$$= 4 \left(\frac{u^3}{3} - \frac{u^5}{5} \right) + C$$

$$= \frac{4}{3} \sin^3 x - \frac{4}{5} \sin^5 x + C$$

b) $\int \frac{\ln x}{x^2} dx \Rightarrow \begin{cases} u = \ln x \\ v = -\frac{1}{x} \\ du = \frac{dx}{x} \\ dv = -\frac{1}{x^2} dx \end{cases}$

$$= -\frac{\ln x}{x} - \frac{1}{x} + C$$

c) $\int_0^{\frac{\pi}{6}} \sqrt{1-x^2} dx \quad x = \sin \theta$

$$= \int_0^{\frac{\pi}{6}} \cos \theta \cdot \cos \theta d\theta$$

$$= \int_0^{\frac{\pi}{6}} \frac{1}{2} (1 + \cos 2\theta) d\theta$$

$$= \left[\frac{\theta}{2} + \frac{\sin 2\theta}{4} \right]_0^{\frac{\pi}{6}}$$

$$= \frac{\pi}{12} + \frac{\sqrt{3}}{8}$$

(4)

d) i) $S_1 = p + q + r = 0$

(7)

ii) $S_2 = p^2 + q^2 + r^2$

$$= (p+q+r)^2 - 2(pq+pr+qr)$$

$$= 0 - 2(k)$$

$$= -2k$$

iii) From $x^3 + kx + 1 = 0$ we

obtain $x^{n+3} + kx^{n+1} + x^n = 0$

Now $p^{n+3} + kp^{n+1} + p^n = 0$,

$$q^{n+3} + kq^{n+1} + q^n = 0$$

and $r^{n+3} + kr^{n+1} + r^n = 0$.

Summing: $S_{n+3} + kS_{n+1} + S_n = 0$ as reqd.

(iv) When $n=0$, $S_0 = 1+1+1 = 3$

and $S_3 + kS_1 + S_0 = 0$

ie $S_3 = -S_0 - kS_1$

$$= -3$$

When $n=2$

$$S_5 + kS_3 + S_2 = 0$$

ie $S_5 = -kS_3 - S_2$

$$= -k(-3) - 2k$$

$$= 5k$$

$$\therefore p^5 + q^5 + r^5 = 5k$$

(3)

Question 3

a) $\int_4^9 \frac{\sqrt{x}}{x-1} dx$ $u^2 = x, u = \sqrt{x}$
 $2u du = dx$

$$= \int_2^3 \frac{2u^2 du}{u^2-1}$$

$$= 2 \int_2^3 \left(\frac{u^2-1}{u^2-1} + \frac{1}{u^2-1} \right) du$$

$$= 2 \int_2^3 \left(1 + \frac{\frac{1}{2}}{u-1} - \frac{\frac{1}{2}}{u+1} \right) du$$

$$= \left[2u + \ln(u-1) - \ln(u+1) \right]_2^3$$

$$= \left(6 + \ln \frac{1}{2} \right) - \left(4 + \ln \frac{1}{3} \right)$$

$$= 2 + \ln \frac{3}{2}$$

(5)

b) i) $x^3 + 2x - 1 = 0$ has roots α, β, δ
 $(-x)^3 + 2(-x) - 1 = 0$ has roots $-\alpha, -\beta, -\delta$

(2)

ie $-x^3 - 2x - 1 = 0$
 $(x^3 + 2x - 1)(-x^3 - 2x - 1) = 0$ has roots $\alpha, -\alpha, \beta, -\beta, \delta, -\delta$.

ii) Since $\alpha + \beta + \delta = 0$
 $\alpha + \beta = -\delta$

(3)

So $\alpha + \beta - 2\delta = -3\delta$.

Similarly $\beta + \delta - 2\alpha = -3\alpha$

and $\alpha + \delta - 2\beta = -3\beta$ and

the equation with these roots is

$$\left(-\frac{x}{3}\right)^3 + 2\left(-\frac{x}{3}\right) - 1 = 0$$

c) i) $(x-1)(1+x+x^2+x^3+x^4)$ (1)

$$= x + x^2 + x^3 + x^4 + x^5$$

$$-1 - x - x^2 - x^3 - x^4$$

$$= x^5 - 1 = P(x).$$

(ii) $P(\alpha) = 0$ and $\alpha \neq 1$ (1)

$$\therefore (\alpha-1)(1+\alpha+\alpha^2+\alpha^3+\alpha^4) = 0$$

Since $\alpha \neq 1, \alpha-1 \neq 0$

$$\therefore 1+\alpha+\alpha^2+\alpha^3+\alpha^4 = 0 \quad \text{--- (1)}$$

(iii) $p+q = \alpha+\alpha^4+\alpha^2+\alpha^3$
 $= -1$ from (1) (1)

and $pq = (\alpha+\alpha^4)(\alpha^2+\alpha^3)$

$$= \alpha^3 + \alpha^4 + \alpha^6 + \alpha^7$$

But $\alpha^6 = \alpha$ and $\alpha^7 = \alpha^2$

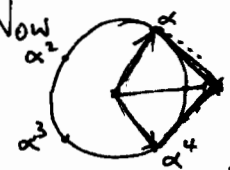
$$\therefore pq = -1 \quad \text{(1)}$$

(iv) p and q are the roots of a quadratic equation in which sum of roots $= -1$ and product of roots $= -1$

ie $x^2 + x - 1 = 0$

$$\therefore x = \frac{-1 \pm \sqrt{1+4}}{2}$$

$$= \frac{-1-\sqrt{5}}{2}, \frac{-1+\sqrt{5}}{2} \quad \text{(1)}$$

Now α^2  p is the greater of the two

$$\therefore p = \frac{-1+\sqrt{5}}{2} \quad \text{(1)}$$

and $q = \frac{-1-\sqrt{5}}{2}$.